

Project Report:

Sinner–Alcaraz on Bluesky:

A Social Network & Sentiment Analysis of the 2025 US Open

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Abstract

This report characterises the online discourse around the tennis rivalry between **Jannik Sinner** and **Carlos Alcaraz** on the **Bluesky** platform during the **2025 US Open** (21 August–10 September 2025). A reproducible pipeline crawls posts from the Bluesky **atproto** API, enriches them with RoBERTa sentiment, dual-backend emotion scoring (NRC lexicon and GoEmotions) and stance signals, builds reply/mention interaction graphs, detects communities (Louvain and Infomap), and renders 20+ visualisations. On a dataset of **9,078 posts** we find a *sparse but highly modular* network (density ≈ 0.002 ; Louvain modularity ≈ 0.97), a *predominantly neutral-to-positive* emotional tone, a sentiment time series that *tracks match outcomes*, and a *sentiment-aware stance classifier* that meaningfully reclassifies users relative to a naive mention-frequency baseline.

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1 Objective

The goal of this project is to **characterise the online discourse** surrounding two of tennis’s biggest rivals—Jannik Sinner and Carlos Alcaraz—on the Bluesky social platform during the 2025 US Open. Concretely the analysis aims to:

- **Describe** the social-network structure built from replies and mentions, identify cohesive communities, and measure user influence via centrality metrics.
- **Quantify** sentiment and emotion expressed in posts using state-of-the-art NLP models (RoBERTa, NRC lexicon, GoEmotions) and compare two emotion backends.
- **Assign a stance** (pro-Sinner, pro-Alcaraz or neutral) to each user by fusing sentiment signals with mention frequency, and evaluate how this refined stance differs from a simpler frequency-only baseline.
- **Visualise** all results to support interpretability—network graphs, sentiment time series, emotion profiles and stance distributions.

The phenomenon under study is the **evolution of fan engagement** during a high-stakes Grand Slam, exploring how rivalry narratives translate into measurable patterns of sentiment, community formation and opinion polarisation.

2 Data

2.1 Collection strategy

Data were collected from Bluesky using its official `atproto` API (`src/crawler.py`). Because the search endpoint does not guarantee exhaustive retrieval over a large window, a **day-by-day** crawling strategy was used:

- For each 24-hour window from 2025-08-21 to 2025-09-10 (UTC), two keyword queries were issued—one **Sinner-oriented** and one **Alcaraz-oriented**—each capped at **5,000 posts/day** to respect rate limits while capturing peak tournament activity.
- A polite sleep was inserted between requests, and automatic retries with exponential backoff handled HTTP 429 rate-limiting.
- Results from both queries were merged and **deduplicated by post URI**.

2.2 Dataset description

After crawling and deduplication the dataset contained **9,078 posts**. The raw schema (`data/sinner_alcaraz_posts.csv`) is summarised in Table 1.

Field	Description
<code>created_at</code>	UTC timestamp
<code>uri</code>	Unique post identifier
<code>text</code>	Original post text
<code>author_handle</code>	User’s handle
<code>author_did</code>	Decentralised identifier
<code>reply_count, like_count, repost_count</code>	Engagement metrics
<code>mentions</code>	List of mentioned user DIDs
<code>links</code>	URLs in the post
<code>reply_parent_did</code>	DID of the user being replied to
<code>query_source</code>	Which query found the post (sinner/alcaraz)

Table 1: Raw data fields collected from Bluesky.

2.3 Preprocessing & enrichment

The raw CSV is transformed into a **frozen** enriched dataset (`data/sinner_alcaraz_processed.csv`, 44 columns) that serves as the single source of truth for all downstream stages. The enrichment (`src/preprocessing.py`) includes:

- **Text cleaning** (HTML unescape, URL/handle removal, normalisation) and **lemmatisation** (spaCy + TweetTokenizer).
- **RoBERTa sentiment** via `cardiffnlp/twitter-roberta-base-sentiment-latest`, producing a `sentiment_category` (positive/neutral/negative) and a continuous `sentiment_compound` = $P(\text{pos}) - P(\text{neg})$.
- **Emotion** with two backends: the **NRC lexicon** (NRCLex) over the 8 Plutchik emotions, and **GoEmotions** (SamLowe/roberta-base-go_emotions), whose 28 fine-grained labels are collapsed onto the same 8 categories for comparison.
- **Named entity recognition** (spaCy) and **entity linking** to detect mentions of Sinner or Alcaraz.
- **Post- and user-level stance** (see Section 3.2).

3 Methodology

The analytical pipeline is a strict DAG (orchestrated by `main.py`).

3.1 Network construction

Nodes are users; edges are reply/mention interactions. Two graphs are built: an **undirected** graph G_u (edge weight = interaction count) and a **directed** graph G_d (author $\rightarrow$ replied/mentioned user). For both we compute degree, closeness and betweenness centrality, in-/out-degree and PageRank. Communities are detected with **Louvain** on G_u and **Infomap** on G_d . **Filtered subgraphs** (connected components ≥ 10 nodes, top communities by post volume) are used for visualisation and stance analysis.

3.2 Stance detection

Post-level stance assigns the RoBERTa compound score to the mentioned player for single-mention posts; dual-mention posts are split via sentence-level sentiment. User-level net stance fuses mean sentiment with mention frequency:

$$\text{net_stance} = 0.50 (\text{mean}_{\text{sinner}} - \text{mean}_{\text{alcaraz}}) + 0.15 (\text{freq}_{\text{sinner}} - \text{freq}_{\text{alcaraz}})$$

(the remaining 0.35 weight is reserved for a future emotion-based component). Users are labelled **sinner** / **alcaraz** / **neutral** with a ± 0.03 threshold. Polarisation is measured via stance assortativity, the cross-stance edge ratio, and the standard deviation of net stance.

4 Results

4.1 Network structure & communities

The interaction graph is **sparse but strongly modular**. Global metrics (Table 2) report a density of ≈ 0.002 and very high community modularity (Louvain ≈ 0.97 , Infomap ≈ 0.94): fans cluster tightly into fanbase-aligned groups rather than mixing. The giant connected component spans only **111 users** ($\approx 16.3\%$), indicating that much of the conversation happens in small, loosely linked clusters.

Metric	Value
Density	0.00203
Transitivity	0.02022
Average clustering	0.01120
Giant connected component (GCC) size	111
GCC fraction of users	0.1625
Louvain modularity	0.9666
Infomap modularity	0.9437

Table 2: Graph-level metrics (`data/network_global_metrics.csv`).

Figure 1 shows the filtered undirected network coloured by fanbase leaning, with node size proportional to degree centrality. High-degree hubs are predominantly news/aggregator accounts (e.g. `tennisupdates`, `theathletic.com`, `thetennisletter`), around which smaller fan clusters of mixed Sinner/Alcaraz/neutral leaning form. The same topology coloured by community and by dominant emotion is shown in Figure 2.



Figure 1: Filtered undirected interaction network coloured by fanbase leaning (orange = Sinner, blue = Alcaraz, grey = neutral); node size $\propto$ degree centrality. Mixed-leaning clusters form around large aggregator/news hubs.



(a) Coloured by Louvain community.

(b) Directed graph, coloured by fanbase.

Figure 2: Filtered network views. Left: community structure on G_u . Right: the directed reply/mention graph G_d coloured by fanbase leaning. Full-resolution and emotion-coloured variants are also generated in `plots/` and `plots/filtered/`.

4.2 Sentiment & emotion profiles

Overall sentiment. The RoBERTa classifier labels the bulk of posts as **neutral (77.2%)**, with **positive (16.2%)** clearly outweighing **negative (6.6%)** (Figure 3). Among posts that do carry sentiment, the conversation skews positive—tennis fans express encouragement and praise far more often than hostility.

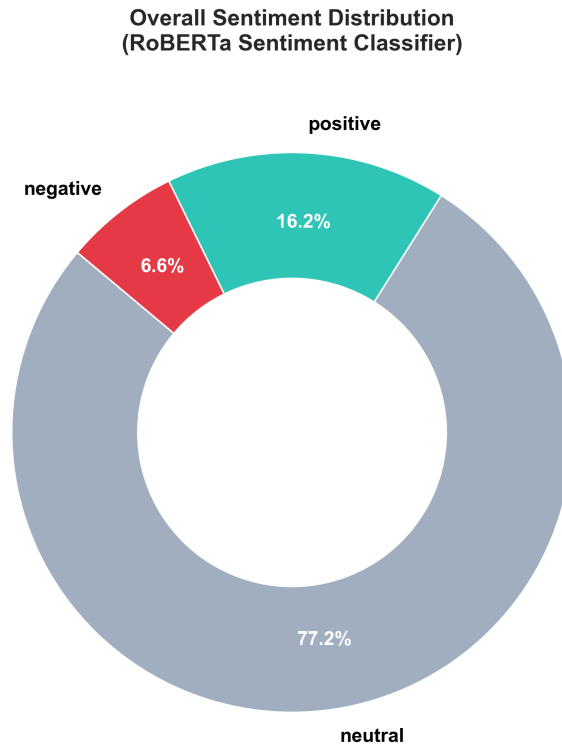


Figure 3: Overall sentiment distribution (RoBERTa). Neutral dominates; positive posts outnumber negative roughly 2.5:1.

Sentiment over time. Figure 4 plots daily average compound sentiment for each fanbase against post volume, annotated with tournament milestones. Sentiment is volatile and reacts to match outcomes: both fanbases swing around the rounds, and the lines **diverge around the final** (7 September, won by Alcaraz), while post volume builds towards the closing weekend.

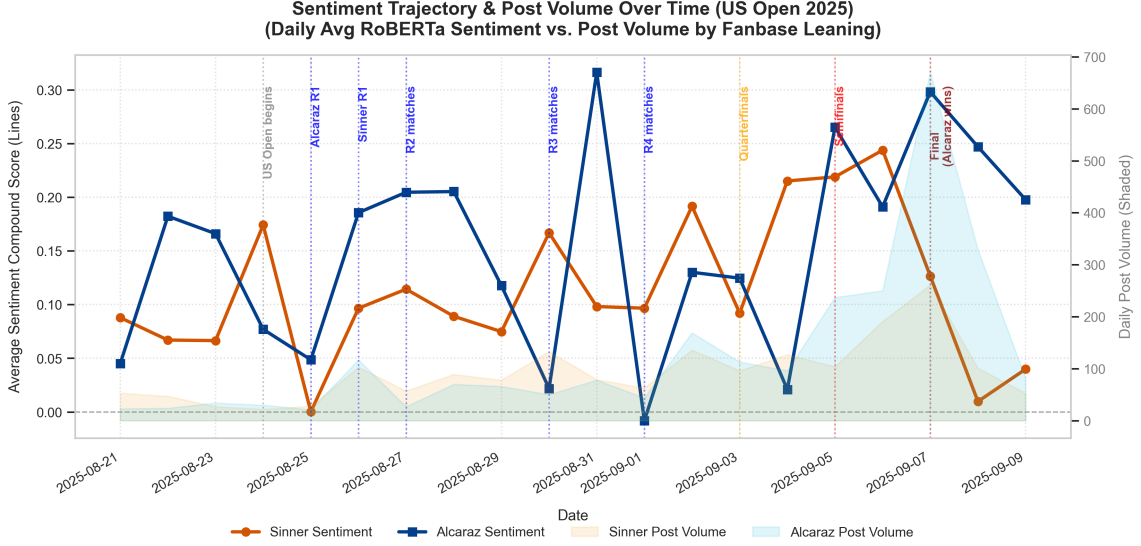


Figure 4: Daily average RoBERTa sentiment for Sinner vs. Alcaraz fans (lines) against daily post volume (shaded), annotated with US Open milestones. Sentiment tracks the tournament and diverges by fanbase around the final.

Emotion backends. Figure 5 compares dominant-emotion shares between the NRC lexicon and GoEmotions (BERT). Both agree that most posts are emotionally *neutral*, but the neural backend is far more conservative (neutral ≈ 0.84 vs. NRC ≈ 0.57). NRC surfaces more *anticipation*, *fear* and *trust*, whereas BERT distributes the remaining mass mainly across *trust*, *anticipation* and *joy*. Crucially, both backends agree that negative emotions (anger, disgust) are rare—consistent with the sentiment donut.

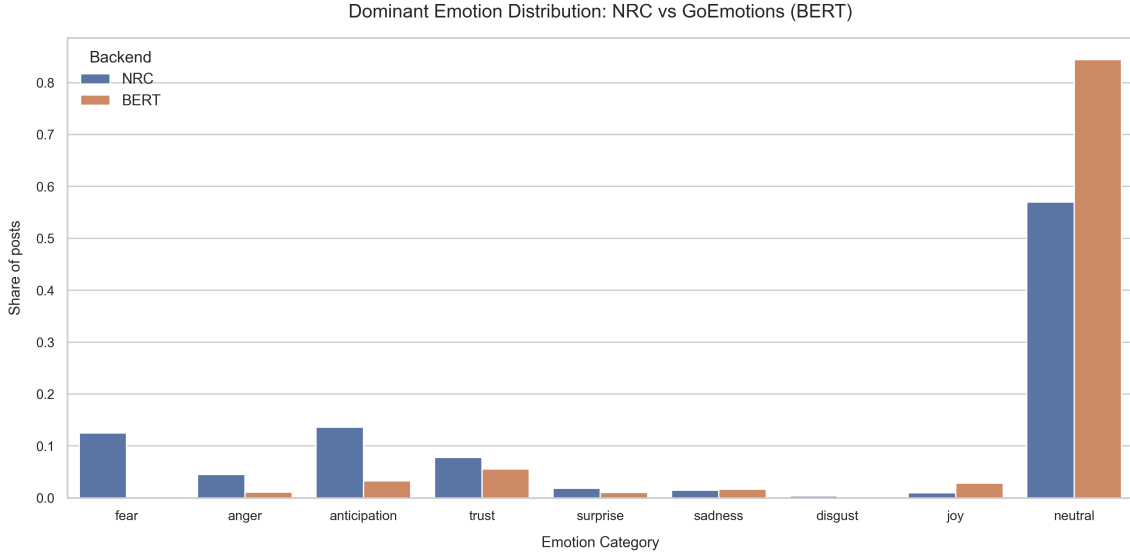


Figure 5: Dominant emotion distribution: NRC lexicon vs. GoEmotions (BERT). The neural backend labels more posts neutral; the lexicon assigns more anticipation/fear/trust.

Community emotion profiles. Per-community emotion profiles (Figure 6) show that the largest Louvain communities share a positive affective signature—*anticipation* and

trust dominate, with *joy* present and negative emotions subdued—but differ in magnitude, reflecting varying engagement intensity. Profiles are generated for both community algorithms (Louvain, Infomap) and both emotion backends (NRC, BERT).

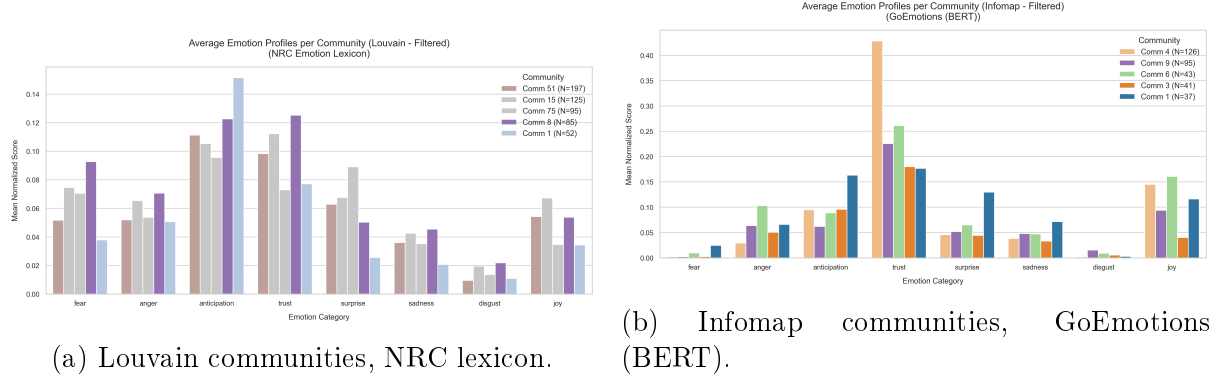


Figure 6: Average emotion profiles for the top communities. Anticipation and trust dominate across clusters; negative emotions stay low.

4.3 Stance & polarisation

The four-panel stance summary (Figure 7) gives the headline picture. Of **2,296 analysed users**, most are **neutral (1,080)**, followed by **Alcaraz (818)** and **Sinner (398)**—a tilt towards Alcaraz consistent with his final victory. Net stance scores are tightly concentrated around zero (score SD = 0.227), and the polarisation diagnostics are mild: stance assortativity is slightly *negative* (−0.047) and the cross-stance edge ratio is 0.108. In other words, despite the high *structural* modularity of the graph, users are *not* strongly segregated by stance—fans of opposing players still interact (often via shared news hubs), so this is not a classic echo chamber.

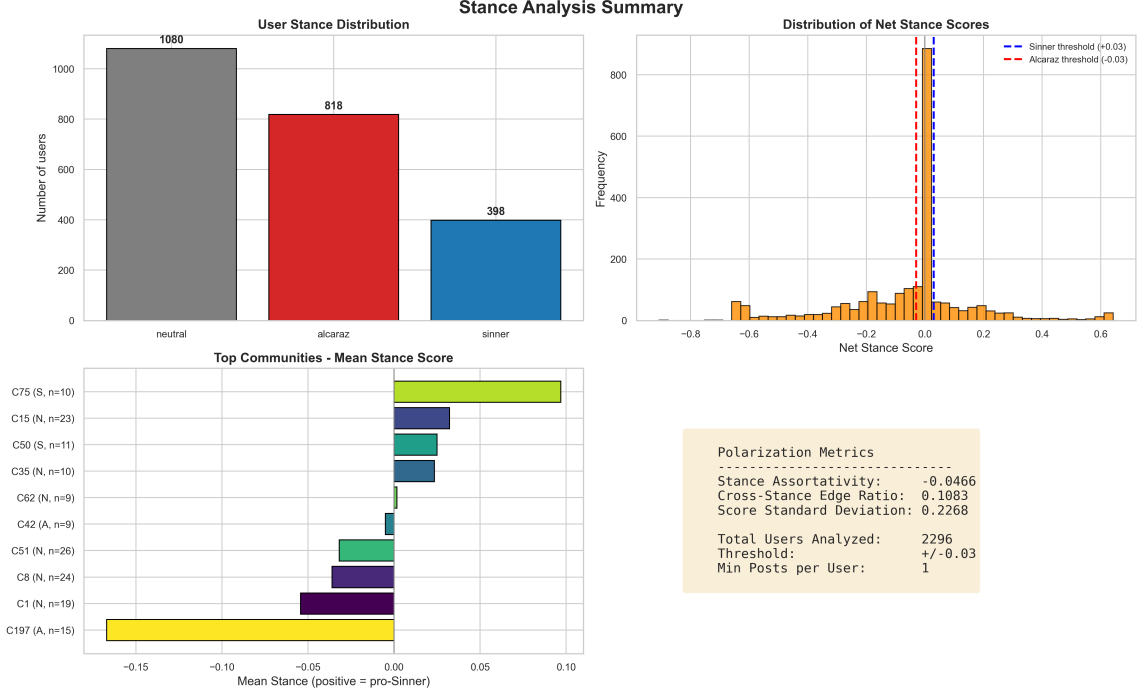


Figure 7: Stance analysis summary. Top-left: user stance counts (neutral 1080, Alcaraz 818, Sinner 398). Top-right: net-stance score histogram with ± 0.03 thresholds. Bottom-left: mean stance of top communities. Bottom-right: polarisation metrics (assortativity -0.047 , cross-stance edge ratio 0.108, score SD 0.227).

Sentiment-aware vs. frequency baseline. Figure 8 contrasts a frequency-only baseline with the refined, sentiment-aware leaning. Folding RoBERTa sentiment into the stance moves users between classes: the Sinner camp shrinks from **454 to 398**, Alcaraz edges up from **814 to 818**, and neutral grows from **1,028 to 1,080**. The reclassified users are typically those who *mention* a player frequently but *express* sentiment inconsistent with that mention pattern (e.g. criticising the player they name most)—exactly the cases a raw mention count mislabels. This demonstrates the value of fusing sentiment with frequency rather than counting mentions alone.

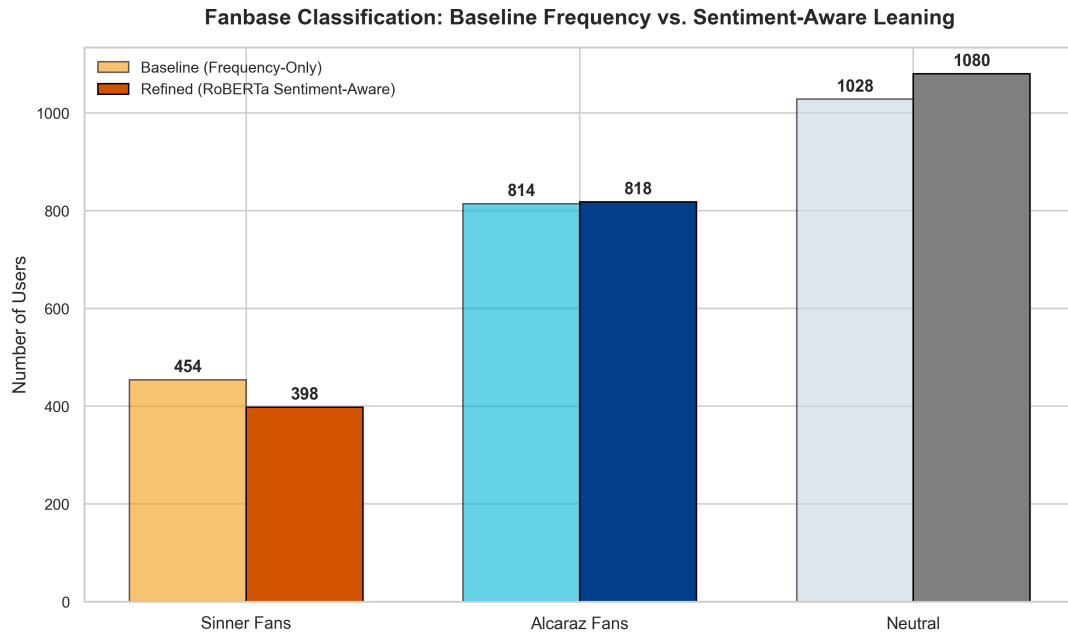


Figure 8: Fanbase classification: frequency-only baseline vs. sentiment-aware refinement. Sentiment reclassifies borderline users, shrinking the Sinner camp and growing the neutral pool.

Community vocabulary. A word cloud over the whole community (Figure 9) is dominated by the players' names and tournament/tennis vocabulary, confirming that the crawl captured on-topic discourse.

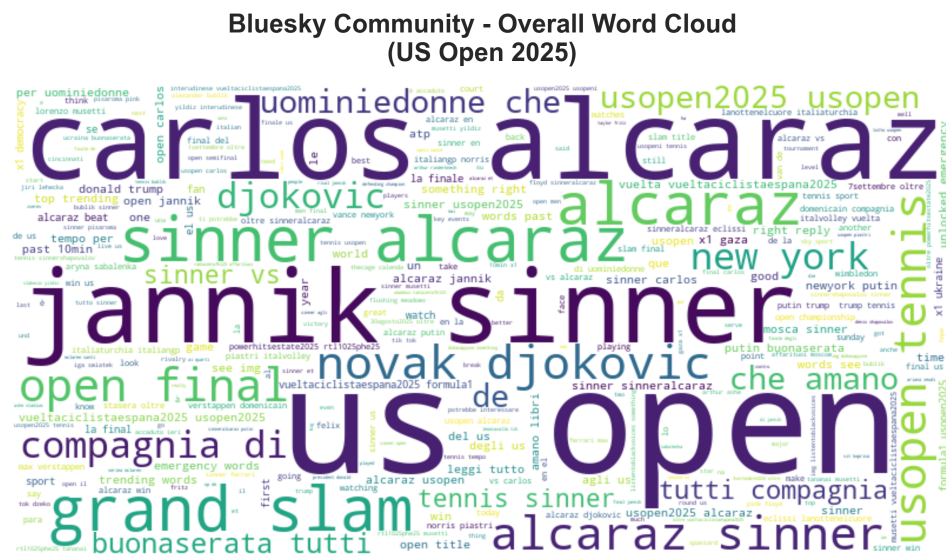


Figure 9: Word cloud of the community corpus—player names and US Open/tennis vocabulary dominate.

5 Discussion

5.1 Interpretation

The analysis paints a picture of a **vibrant, on-topic, but only mildly polarised** online community:

- **Structurally fragmented, not opinion-segregated.** The graph is extremely modular (Louvain $Q \approx 0.97$) yet stance assortativity is near zero. Clustering is driven by *who replies to whom* (frequently shared news/aggregator hubs) more than by partisan allegiance.
- **Positive, low-toxicity tone.** Most posts are neutral; among sentiment-bearing posts, positive outweighs negative $\approx 2.5:1$, and negative emotions (anger, disgust) are rare under both emotion backends.
- **Sentiment tracks the tournament.** The time series reacts to rounds and diverges by fanbase around the final, illustrating how match outcomes drive emotional response.
- **Sentiment-aware stance matters.** Refining a frequency baseline with sentiment reclassifies a meaningful slice of users, correcting cases where mention frequency and expressed sentiment disagree.

5.2 Strengths

- **Reproducible DAG** with a frozen intermediate dataset, so any downstream stage can be re-run without recomputing the expensive NLP enrichment (seeds fixed at 42).
- **Two emotion backends** (lexicon vs. neural) provide robustness and expose how sensitive emotion attribution is to the scoring method.
- **Fine-grained stance** via sentence-level splitting of dual-mention posts.

5.3 Limitations & future work

- **Recall.** The Bluesky search API does not guarantee exhaustive retrieval; posts without the chosen keywords are missed.
- **Sarcasm & irony.** RoBERTa sentiment can misread sarcastic praise/criticism; sarcasm detection is a natural extension.
- **Weight calibration.** The stance weights (0.50, 0.15, with 0.35 reserved) are heuristic; a data-driven fit against labelled stances would improve accuracy and let the reserved emotion component be activated.
- **Static communities.** Community detection is computed once over the whole window; a temporal (sliding-window) analysis could reveal how clusters form and dissolve around matches.

5.4 Conclusion

By integrating social-network analysis, NLP-based sentiment/emotion mining and sentiment-aware stance detection, this project characterises the Bluesky discourse around the Sinner–Alcaraz rivalry at the 2025 US Open. The community is sparse and highly modular yet not opinion-segregated, its tone is neutral-to-positive, its sentiment tracks match outcomes, and a sentiment-aware stance classifier improves on a naive frequency baseline. The pipeline is modular and reproducible, and readily adapts to other rivalries or platforms.